Mobile Application Development – CA1: Review 1
Idea & Proposal Review

Milestone: Problem definition, concept, and project scope

Section	Details
Group Number	*11*
Team Members	*1. JOYLYN PRINCITA FERNANDES — 25MCAR0099* *2. SYED FAIZAN PASHA — 25MCAR0138* *3. VISHWAJEET SINGH — 25MCAR0219*
Title	*AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application*
App Name	*MentalCare AI*
Problem Statement	*Stress is a common real-life issue among students, employees, and working professionals. Academic pressure, workload, poor sleep, deadlines, and personal responsibilities can cause increasing stress and mental fatigue. People may not recognize these patterns early. Existing wellness applications often provide generic content rather than analyzing a user's own words and daily patterns. AI MentalCare aims to provide an early-warning and wellness-support mechanism through real-time AI analysis.*
Abstract	*AI MentalCare is a real-time AI-powered mobile application that helps users monitor stress and possible burnout patterns. Users can complete daily mental check-ins, write journal entries, and record wellness information such as sleep, activity, mood, and workload. Natural Language Processing (NLP) and Machine Learning are used to classify stress-related text and estimate a stress level. The system also tracks stress scores over time to identify increasing patterns and provides personalized wellness recommendations and timely alerts. The application is intended as a wellness and early-warning tool, not a medical diagnostic system.*
Target Users	*Men and women, particularly students, employees, and working professionals who experience academic, workplace, lifestyle, or day-to-day stress and want a simple way to monitor their mental wellness and recognize increasing stress patterns.*
Planned Core Features	*1. AI Mental Check-in: Analyze user-written text using NLP and classify stress level.* *2. Real-Time Stress Monitoring: Update the user's stress status and dashboard after each check-in.* *3. Burnout Trend Detection: Analyze multi-day stress patterns and identify persistent/increasing stress.* *4. Personalized Recommendations: Suggest wellness actions based on the user's current pattern.* *5. Real-Time Alerts: Send notifications when elevated or continuously increasing stress indicators are detected.*
App Icon	

Remarks:

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Vishwajeet Singh
Joylyn.

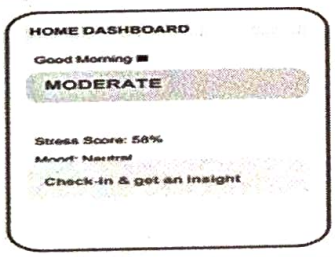
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Signature of Faculty In charge
Mr. Aldrin Jose Antony

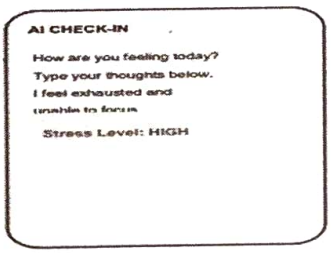
Mobile Application Development – CA1: Review 2
UI/UX Design Review

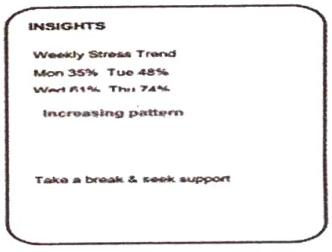
Milestone: Wireframes, mockups, and visual design language

Section	Details
Group Number	*11*
Team Members	*1. JOYLYN PRINCITA FERNANDES — 25MCAR0099* *2. SYED FAIZAN PASHA — 25MCAR0138* *3. VISHWAJEET SINGH — 25MCAR0219*
Title	*AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application*
App Name	*MentalCare AI*
Screens Designed	*Splash Screen, Login, Registration, Home Dashboard, AI Mental Check-in, AI Chat, Stress Result, Stress History, Burnout Analysis, Personalized Recommendations, Notifications, Profile & Settings, Trusted Support/Emergency Contact.*
Wireframe / Mockup Tool	*Figma — wireframes and visual mockups are planned in Figma. The sample wireframes below represent the intended design direction for the review.*
Colour Palette & Typography	*Primary Green: #10B981 Dark Green: #047857 Mint Background: #ECFDF5 AI Accent: #8B5CF6 Moderate: #F59E0B High: #EF4444. Typography: clean, highly readable sans-serif typography such as Roboto. Green represents calmness, wellness and growth; purple is reserved for AI-related actions.*
Design Consistency Notes	*Consistent green primary actions, common card styles, repeated navigation, readable typography, clear status labels, sufficient contrast, and simple icons will be used across screens. Stress levels will use both colour and text labels so that information is not communicated by colour alone.*
Layout & Visual Balance	*The interface will use a clean single-column mobile layout with consistent margins, spacing and alignment. Important information such as the current stress level and AI recommendation will receive visual priority. Cards and sections will be kept short to reduce cognitive load.*
Add-ons as per Remarks	*Initial design includes AI check-in, stress trend visualization, burnout warning, personalized recommendations, real-time notifications, and trusted-support options. Further changes will be incorporated based on faculty remarks*

Sample Wireframes / Mockups







Evaluation – UI Design & Visual Appeal

Remarks:

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Mobile Application Development – CA1: Review 3

Development & Responsiveness Review

Milestone: Core build progress and responsive layout implementation

Section	Details
Group Number	*11*
Team Members	*1. JOYLYN PRINCITA FERNANDES — 25MCAR0099* *2. SYED FAIZAN PASHA — 25MCAR0138* *3. VISHWAJEET SINGH — 25MCAR0219*
Title	*AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application*
App Name	*MentalCare AI*
Features Implemented	*Initial Android application structure; Login and Registration screens; Home Dashboard; AI Mental Check-in and journal input; Stress Result screen; stress history/trend interface; personalized recommendation section; Firebase integration structure; and responsive layouts for different Android screen sizes.*
Tech Stack Used	*Android Studio, Kotlin, XML/Material Design, Firebase Authentication, Firebase Firestore/Realtime Database, Firebase Cloud Messaging, Python, Flask/FastAPI, Scikit-learn, NLP, REST API and Room Database.*
Devices / Screen Sizes Tested	*Android Studio emulator using compact and standard Android phone profiles. Small, medium and large phone layouts and portrait orientation are considered for responsiveness testing.*
Responsive Design Approach	*Constraint-based layouts, ScrollView for longer content, dp/sp dimensions, flexible cards, scalable text and consistent margins are used so the application adapts to different screen sizes without clipping or horizontal overflow.*
Known Issues / Bugs	*End-to-end AI backend integration, notification testing and some edge/error states still require final verification. Minor UI spacing and responsiveness issues may be refined before the next review.*
Add-ons as per Remarks	*Improve loading/error states, stress trend visualization, notification handling, accessibility/readability and AI status indicators based on faculty feedback.*

Evaluation – Responsiveness & Interactivity:

Remarks:

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Mobile Application Development – CA1: Review 4
UX, Navigation & Usability Testing

Milestone: User flow, interactivity, and usability feedback

Section	Details
Group Number	*11*
Team Members	*1. JOYLYN PRINCITA FERNANDES — 25MCAR0099* *2. SYED FAIZAN PASHA — 25MCAR0138* *3. VISHWAJEET SINGH — 25MCAR0219*
Title	*AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application*
App Name	*MentalCare AI*
Navigation Flow Summary	*Splash Screen □ Login/Register □ Home Dashboard □ AI Mental Check-in □ AI Analysis/Stress Result □ Recommendations. Bottom navigation provides quick access to Home, AI Check-in, Insights/History, Notifications and Profile. Back navigation returns to the previous logical screen.*
Interactive Components Used	*Buttons, text fields, password fields, mood selectors, cards, lists/RecyclerView, bottom navigation, dialogs, notification controls, progress/status indicators, date/time inputs and scrollable content.*
Usability Testing Method	*Peer testing with the three team members and selected student users. Testers perform common tasks such as login, daily check-in, viewing the AI result, opening stress history, navigating between screens and viewing recommendations.*
Usability Feedback Received	*The interface should keep navigation simple, make the AI check-in prominent, improve readability of stress levels, maintain the green visual theme, provide clear loading/error messages and avoid overcrowding the dashboard.*
Navigation Improvements Planned	*Reduce unnecessary transitions, keep important actions within one or two taps, improve back navigation, retain bottom-navigation state, improve empty/error states and simplify AI result and recommendation screens.*
Add-ons as per Remarks	*Add onboarding/help text, accessibility labels, confirmation dialogs where needed, clearer notification messages and a visible disclaimer that the application is not a medical diagnostic tool.*

Evaluation – User Experience & Navigation

Remarks:

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Mobile Application Development – CA1: Review 5

Final Presentation & Deployment Review

Milestone: Completed app, deployment, and final demo

Section	Details
Group Number	*11*
Team Members	*1. JOYLYN PRINCITA FERNANDES — 25MCAR0099* *2. SYED FAIZAN PASHA — 25MCAR0138* *3. VISHWAJEET SINGH — 25MCAR0219*
Title	*AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application*
App Name	*MentalCare AI*
Deployment Details	*Final Android APK will be generated from Android Studio for demonstration and testing. The AI backend will be hosted on a suitable Python server, with Firebase used for authentication, cloud synchronization and notifications. Final APK/repository/hosting links will be added before submission.*
Final Feature Summary	*User authentication; mental check-ins and journal input; NLP-based stress classification; Low/Moderate/High stress result; multi-day stress trend analysis; possible burnout-risk pattern warning; personalized wellness recommendations; Firebase cloud data; real-time dashboard updates; notifications; stress history; profile; trusted-support option; responsive UI and wellness disclaimer.*
Demo Notes	*Live demo flow: Login ☐ Dashboard ☐ AI Mental Check-in ☐ enter sample text ☐ AI analysis ☐ stress classification ☐ save result ☐ stress history/trend ☐ increasing-stress warning ☐ personalized recommendation ☐ notification/support option. Responsive layout will also be demonstrated.*
Project Report Submitted	*Yes — final report will include project overview, problem statement, objectives, architecture, UI/UX, AI/NLP workflow, database design, implementation, testing, screenshots, limitations and future enhancements. Final links will be attached according to faculty instructions.*
App Icon	*Brain + Leaf + Heart + AI concept using green/mint as the primary colours with a small purple AI accent.*
Add-ons as per Remarks	*Incorporate final faculty-requested UI/UX changes, additional test cases, improved error handling, final AI model evaluation, deployment verification, documentation updates and final presentation preparation.*

Final Evaluation – CA1 Rubric Total 15 Marks

Remarks:

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Mobile Application Development – CA3: Review 1

Idea & AI Use-Case Proposal

Milestone: Problem definition, AI/ML use case, and project scope

Section	Details
Group Number	11
Team Members	Joylyn Princita Fernandes (25MCAR0099); Syed Faizan Pasha (25MCAR0138); Vishwajeet Singh (25MCAR0219)
Title	AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application
App Name	MentalCare AI
Proposed AI/ML Feature	Stress-level classification (Low / Moderate / High) from user check-in text, with personalized wellness recommendations.
Why AI is Needed	AI/NLP can interpret free-form user text and identify stress patterns more consistently than fixed rule-only inputs, enabling adaptive classification and recommendations.
Target Users	Students and young adults who want a simple, private wellness check-in, stress insight, journaling and self-care support tool.
Add-ons as per Remarks	Implemented as a complete working product: AI stress check-in, journal, insights/history, wellness recommendations, chatbot, lifestyle tools, safety support, Firebase authentication/data, notifications and responsive UI.

Remarks:

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Mobile Application Development – CA3: Review 2

AI/ML Model & Data Review

Milestone: Dataset preparation, model selection, and training approach

Section	Details
Group Number	11
Team Members	Joylyn Princita Fernandes (25MCAR0099); Syed Faizan Pasha (25MCAR0138); Vishwajeet Singh (25MCAR0219)
Title	AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application
App Name	MentalCare AI
Dataset Used	A labelled text dataset for stress/mental-wellness classification was used for the trained NLP component. The exact source and sample count are not stated in the supplied CA3 template/project notes.
Data Preprocessing	Text cleaning and normalization, tokenization/vectorization as required by the trained classifier, label preparation, and conversion of user check-in text into model-ready input.
Model / Algorithm Chosen	A trained NLP text-classification model exposed through the project's Python REST backend. The classifier was selected to map free-form check-in text to Low, Moderate or High stress categories.
Training & Validation Approach	Training/validation was performed on labelled text data and checked using classification metrics. Exact split and validation figures are not stated in the supplied CA3 template/project notes.
Model Performance (so far)	Functional final output: Low / Moderate / High stress classification with result history and recommendations. A verified numerical accuracy/precision/recall value is not stated in the supplied materials, so no value is invented here.
Integration Plan	Flutter mobile app sends the check-in securely over HTTPS to the existing Python Flask/Fast API REST API; the backend preprocesses the text, runs the trained classifier, returns the result, and the app stores/displays the result using the existing Firebase architecture.
Add-ons as per Remarks	Completed integration and refinements were implemented, including history, trend handling, personalized recommendations, reliable error states and production-oriented authentication/data flow.

Evaluation – AI Integration & Performance (partial)

Remarks:

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Mobile Application Development – CA3: Review 3

App Development & Functionality Review

Milestone: Core app build and functional integration of the AI feature

Section	Details
Group Number	11
Team Members	Joylyn Princita Fernandes (25MCAR0099); Syed Faizan Pasha (25MCAR0138); Vishwajeet Singh (25MCAR0219)
Title	AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application
App Name	MentalCare AI
Features Implemented	Authentication and account setup; Home/Dashboard; AI Mental Check-in; stress result; stress history/trends; journal; AI wellness chatbot and chat history; daily wellness quiz; lifestyle/sleep/habits/goals; recommendations; notifications/reminders; profile; safety/SOS and trusted contacts; support; About; responsive navigation; Firebase integration.
Tech Stack Used	Flutter/Dart mobile application; Firebase Authentication and database services; Python Flask/Fast API REST backend; NLP/ML text classifier; HTTPS REST communication; Android release build tooling.
AI Feature Functionality	User enters a free-form wellness/check-in message → Flutter sends it to the Python backend → text is preprocessed → trained NLP classifier predicts Low/Moderate/High stress → result and personalized recommendations are returned → result/history are shown in the app and persisted through the existing backend/Firebase flow.
Error Handling / Edge Cases	Handles empty/invalid input, authentication failures, network/API failures and unavailable results with user-friendly validation, loading/error states and retry paths. The wellness output is presented as supportive information, not a medical diagnosis.
Known Issues / Bugs	No critical functional issue is recorded for the completed build. Release verification includes authentication, AI response flow, history, navigation, notifications, safety actions and Android packaging.
Add-ons as per Remarks	Final refinements were implemented for compact dashboard design, modern navigation icons, profile photo support, daily quiz uniqueness, chatbot reliability, launcher icon configuration and Play-Store-oriented UI polish.

Evaluation – Application Functionality

Remarks:

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Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 4

UI/UX & AI Output Presentation Review

Milestone: Interface design and clarity of AI-generated results

Section	Details
Group Number	11
Team Members	Joylyn Princita Fernandes (25MCAR0099); Syed Faizan Pasha (25MCAR0138); Vishwajeet Singh (25MCAR0219)
Title	AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application
App Name	MentalCare AI
Screens Designed	Splash; Login; Registration; Email verification; Phone OTP; Account Setup; Dashboard; AI Mental Check-in; Stress Result; Insights/History/Trend; Journal; AI Chatbot; Daily Wellness Quiz; Lifestyle/Wellness; Sleep; Goals/Habits; Recommendations; Notifications; My Profile; Safety & Emergency; Support; About; Settings.
Navigation Flow	Authentication → Account Setup → Dashboard. From the dashboard users can start an AI check-in, open Insights/History, Journal, Lifestyle or Support, while the AI chatbot is available as a floating assistant. Each major feature has its own screen rather than being duplicated on the dashboard.
AI Output Presentation	AI results are shown with clear Low/Moderate/High labels, concise stress insight, history/trend views and actionable wellness recommendations. The interface avoids unsupported medical claims or fabricated confidence values.
Interactive Components Used	Material-style buttons, forms, text fields, cards, lists, dialogs, OTP input, progress/loading states, charts/trend views, tabs/navigation, profile image picker, reminder controls and floating AI assistant action.
Responsiveness Check	Responsive Flutter layouts were designed for common Android phone screen sizes and portrait use, with adaptive spacing, scrollable content and safe-area handling. Final release verification is performed on the target Android device/emulator.
Add-ons as per Remarks	UI/UX was refined into a minimal, premium wellness style using green/mint branding, rounded cards, modern rounded icons, compact dashboard content and a consistent AI assistant presentation.

Evaluation – UI/UX Design

Remarks:

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Mobile Application Development – CA3: Review 5

Final Deployment & Demo Review

Milestone: Deployed application, final AI performance, and live demo

Section	Details
Group Number	11
Team Members	Joylyn Princita Fernandes (25MCAR0099); Syed Faizan Pasha (25MCAR0138); Vishwajeet Singh (25MCAR0219)
Title	AI MentalCare – Real-Time AI-Powered Stress & Burnout Detection Mobile Application
App Name	MentalCare AI
Deployment Details	Android release build (APK/AAB) with the existing Python REST backend and Firebase services. The project is prepared for Android deployment; no public Play Store URL or APK download URL was supplied in the available project materials.
Final AI Model Performance	Final functional validation confirms the trained NLP stress-classification feature and Low/Moderate/High output. Exact final numerical model metrics are not present in the supplied template/project notes and are therefore not fabricated.
Final Feature Summary	Completed build includes authentication and account setup, AI stress check-in/classification, stress history and trends, journal, contextual AI chatbot, daily wellness quiz, personalized recommendations, lifestyle/sleep/goals/habits, reminders/notifications, profile, safety/SOS, trusted support, Firebase persistence and responsive modern UI.
Demo Notes	Live demo: registration/login and account setup → dashboard → AI mental check-in → stress result and recommendation → history/trend → journal → AI chatbot conversation → wellness quiz → lifestyle/wellness → profile → safety/SOS → notification/reminder flow.
Project Report Submitted	Yes – final project documentation prepared.
Add-ons as per Remarks	Final refinements and release checks completed. The build is intended to demonstrate the complete AI-enabled mobile application, its backend integration, user flows and wellness-focused UI.

Final Evaluation – CA3 Rubric (Total 15 Marks)

Syed Faizan Pasha
Vishwajeet Singh
Joylyn

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